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Task:

1 use db first approach in entity framework and list the trees/products in the homepage

Solution :

```
using System;
using System.Collections.Generic;

namespace Day6DBFirst.Models;

public partial class Tree
{
    public int Id { get; set; }

    public string Name { get; set; } = null!;

    public string Description { get; set; } = null!;

    public int NoOfTree { get; set; }
}

using System.Diagnostics;
using Day6DBFirst.Data;
using Day6DBFirst.Models;
using Microsoft.AspNetCore.Mvc;

namespace Day6DBFirst.Controllers
{
    public class HomeController : Controller
    {
        private readonly ILogger<HomeController> _logger;

        public TrainingDbContext train;

        public HomeController(ILogger<HomeController> logger, TrainingDbContext train)
        {
            _logger = logger;
            this.train = train;
        }
    }
}
```

```

    public IActionResult Index()
    {
        var alltree = train.Trees.ToList();
        return View(alltree);
    }

    public IActionResult Privacy()
    {
        return View();
    }

    [ResponseCache(Duration = 0, Location = ResponseCacheLocation.None, NoStore =
true)]
    public IActionResult Error()
    {
        return View(new ErrorViewModel { RequestId = Activity.Current?.Id ??
HttpContext.TraceIdentifier });
    }
}

```

```

using System;
using System.Collections.Generic;
using Day6DBFirst.Models;
using Microsoft.EntityFrameworkCore;

```

```

namespace Day6DBFirst.Data;

```

```

public partial class TrainingDbContext : DbContext
{
    public TrainingDbContext()
    {
    }

    public TrainingDbContext(DbContextOptions<TrainingDbContext> options)
        : base(options)
    {
    }

    public virtual DbSet<Product> Products { get; set; }

    public virtual DbSet<Student> Students { get; set; }

    public virtual DbSet<Tree> Trees { get; set; }
}

```

protected override void OnConfiguring(DbContextOptionsBuilder optionsBuilder)
#warning To protect potentially sensitive information in your connection string, you should move it out of source code. You can avoid scaffolding the connection string by using the Name= syntax to read it from configuration - see <https://go.microsoft.com/fwlink/?linkid=2131148>. For more guidance on storing connection strings, see <https://go.microsoft.com/fwlink/?LinkId=723263>.

```
=> optionsBuilder.UseSqlServer("Data Source=PTPLL614;Initial  
Catalog=TrainingDB;Integrated Security=true;TrustServerCertificate=True;");
```

```
protected override void OnModelCreating(ModelBuilder modelBuilder)  
{  
    modelBuilder.Entity<Product>(entity =>  
    {  
        entity.ToTable("Product");  
  
        entity.Property(e => e.Id).HasColumnName("id");  
        entity.Property(e => e.Amount).HasColumnName("amount");  
        entity.Property(e => e.Description).HasColumnName("description");  
        entity.Property(e => e.Name).HasColumnName("name");  
    });  
  
    modelBuilder.Entity<Student>(entity =>  
    {  
        entity  
            .HasNoKey()  
            .ToTable("Student");  
  
        entity.Property(e => e.CustomerId).HasColumnName("customer_id");  
        entity.Property(e => e.Location)  
            .HasMaxLength(50)  
            .IsUnicode(false)  
            .HasColumnName("location");  
        entity.Property(e => e.Name)  
            .HasMaxLength(50)  
            .IsUnicode(false);  
    });  
  
    modelBuilder.Entity<Tree>(entity =>  
    {  
        entity.ToTable("tree");  
  
        entity.Property(e => e.Id).HasColumnName("id");  
        entity.Property(e => e.Description).HasColumnName("description");
```

```

        entity.Property(e => e.Name)
            .HasMaxLength(300)
            .HasColumnName("name");
        entity.Property(e => e.NoOfTree).HasColumnName("no_of_tree");
    });

    OnModelCreatingPartial(modelBuilder);
}

partial void OnModelCreatingPartial(ModelBuilder modelBuilder);
}

using Day6DBFirst.Data;
using Day6DBFirst.Models;

var builder = WebApplication.CreateBuilder(args);
builder.Services.AddDbContext<TrainingDbContext>();

// Add services to the container.
builder.Services.AddControllersWithViews();

var app = builder.Build();

const string connection = "Data Source=PTPLL614;" + "Initial Catalog=TrainingDB;"
    + "Integrated Security=true;" + "TrustServerCertificate=True";

// Configure the HTTP request pipeline.
if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Home/Error");
    // The default HSTS value is 30 days. You may want to change this for production scenarios,
    see https://aka.ms/aspnetcore-hsts.
    app.UseHsts();
}

app.UseHttpsRedirection();
app.UseStaticFiles();

app.UseRouting();

app.UseAuthorization();

```

```
app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}");
```

```
app.Run();
```

```
@{
    ViewData["Title"] = "Home Page";
}
```

```
@model List<Tree>
```

```
<div class="container mt-4">
```

```
    <h1 class="text-center text-success mb-4"> Welcome To Tree Page </h1>
```

```
    <div class="table-responsive">
```

```
        <table class="table table-bordered table-striped table-hover text-center align-middle">
```

```
            <thead class="table-dark">
```

```
                <tr>
```

```
                    <th scope="col">#</th>
```

```
                    <th scope="col">Tree Name</th>
```

```
                    <th scope="col">Description</th>
```

```
                    <th scope="col">No. of Trees</th>
```

```
                </tr>
```

```
            </thead>
```

```
            <tbody>
```

```
                @if (Model != null && Model.Any())
```

```
                {
```

```
                    int i = 1;
```

```
                    foreach (var tree in Model)
```

```
                    {
```

```
                        <tr>
```

```
                            <td>@i</td>
```

```
                            <td>@tree.Name</td>
```

```
                            <td>@(string.IsNullOrEmpty(tree.Description) ? "No Description" :
tree.Description)</td>
```

```
                            <td>@tree.NoOfTree</td>
```

```
                        </tr>
```

```
                        i++;
```

```
                    }
```

```
                }
```

```
            else
```

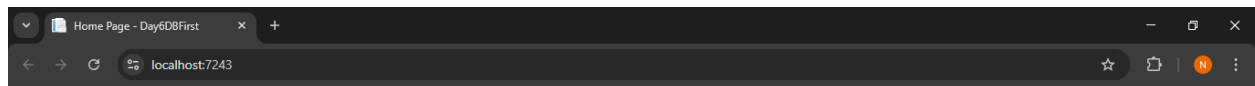
```
            {
```

```

        <tr>
          <td colspan="4" class="text-muted">No trees found 🌱 </td>
        </tr>
      }
    </tbody>
  </table>
</div>
</div>

```

Output:



Day6DBFirst Home Privacy

Welcome To Tree Page

#	Tree Name	Description	No. of Trees
1	mango	Seasonal Fruit	35
2	Neem	Medicinal Value	26
3	JackFruit	Tasty	34
4	Banana	Healthy	20